

AHMED MOHAMED AMIN

+20 128 802 4972 | Giza, Egypt | ahmed.m.amin2003@gmail.com

[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [IEEE Publication](#)

EDUCATION

African Institute for Mathematical Sciences (AIMS)

Cape Town, South Africa

M.Sc. Mathematical Sciences — AI for Science | Google DeepMind Scholarship (full funding) Sept 2026 – July 2027

Zewail City of Science and Technology

Giza, Egypt

B.Sc. Communication & Information Engineering (ABET) | GPA: 3.64/4.0, Cum Laude Sept 2021 – June 2026

- **Coursework:** Machine Learning, Deep Learning, NLP, Computer Vision, Big Data Analytics, Statistical Inference, Algorithms & Data Structures, Linear Algebra.

EXPERIENCE

Takhail | AI Engineer | Giza, Egypt

Dec 2025 – Present

- Built and shipped **four bilingual (Arabic/English) AI assistants** for paying clients, owning each system from API design through deployment.
- Cut inference cost per message by roughly $5\times$ (about 5 EGP down to 1 EGP) by redesigning the embedding and retrieval layer.
- Raised answer quality using hybrid BM25 and vector retrieval with RRF fusion and MMR reranking, plus query routing that selects the right knowledge source for each question.
- Designed an **asynchronous architecture** on Celery that runs speech-to-text, text-to-speech, and retrieval in parallel, so one API serves both voice and text conversations.
- Deployed containerised microservices (FastAPI, ChromaDB, MongoDB, Redis, MinIO) on AWS EC2 with Docker Swarm and zero-downtime releases, monitored through a Promtail/Loki tracing dashboard.
- **Largest systems:** an Arabic advisory chatbot over a 23,512-chunk corpus; a property-sales assistant with a 702-image catalogue and a 66-case evaluation harness; and a trading assistant running on a 9-container stack.

PUBLICATION

IEEE CSCloud 2025 — 12th Int. Conference on Cyber Security and Cloud Computing

Aug 2025

- L. Rana, **A. M. Amin**, H. Ameer, and S. Latif, “FinVerse Agent: Resource-Efficient Financial LLMs with Modular LoRA Fine-Tuning and Task Routing,” in *Proc. IEEE CSCloud*, 2025.
- **My contribution:** second author, and the Zewail City side of a collaboration with NUST Islamabad. Fine-tuned 15 task-specific LoRA adapters on a 4-bit quantised LLaMA-3.1-8B ($10\times$ cheaper than full fine-tuning) and built the Planner–Executor–Verifier loop that picks an adapter per task and validates its output.
- **Results:** **93.07%** on CFA Level I (GPT-4o: 88.10%) and **97.24 F1** on financial NER (BloombergGPT: 60.82), using 75% less memory. **Won 1st Place, FinAI Contest 2025 (Task 2).**

GRADUATION PROJECT

Rafeeq — AI Retrofit Kit for Manual Wheelchairs | Arabic NLP, Robotics, Edge AI

Sept 2025 – 2026

- Designed a kit that adds autonomous navigation, Arabic voice control, and health monitoring to any manual wheelchair at **60–70% below** the cost of an imported smart wheelchair.

- Built an offline Egyptian Arabic speech pipeline (faster-whisper INT8) on a Raspberry Pi 5, reaching **87% command accuracy at 0.9s latency** with no internet connection.
- Added indoor navigation using LiDAR and depth-camera SLAM (ROS 2 Nav2), with a three-tier controller that converts commands into motor movement in under 20 ms.
- Awarded a **\$1,600 grant** from Egypt’s Ministry of Higher Education, selected nationally across all Egyptian universities.

TECHNICAL SKILLS

Languages: Python, C++, SQL

AI & NLP: RAG, Hybrid Retrieval (BM25 + Vector), Reranking, PyTorch, Hugging Face Transformers, LangChain, LoRA/QLoRA, PEFT, 4-bit Quantisation, vLLM, Whisper, Computer Vision

Data: PostgreSQL/pgvector, ChromaDB, FAISS, MongoDB, Redis

Infrastructure: AWS EC2, Docker, Docker Swarm, Celery, Promtail/Loki, Linux, Git, CI/CD

Software Engineering: FastAPI, REST API design, microservices, system design, ROS 2

Spoken Languages: Arabic (native), English (professional — **IELTS Academic 6.5**, CEFR B2)

AWARDS & LEADERSHIP

Google DeepMind Scholarship (2026) — full funding for the AI for Science M.Sc. at AIMS South Africa.

Full Excellence Scholarship (2021–2026) — five years of full tuition, Zewail City; Provost’s Honors Roll, 7 semesters.

Egypt Scholars, Basic Lab 12 (2026) — selected for a 3-month program in research methods and scientific writing.

Elected Student Representative, Zewail City (2021–2025) — represented four CIE cohorts, over 200 students, for four consecutive years.

Teaching Assistant, Math 101, Zewail City (2022–2023) — led tutorials for 50+ first-year students in calculus and linear algebra.